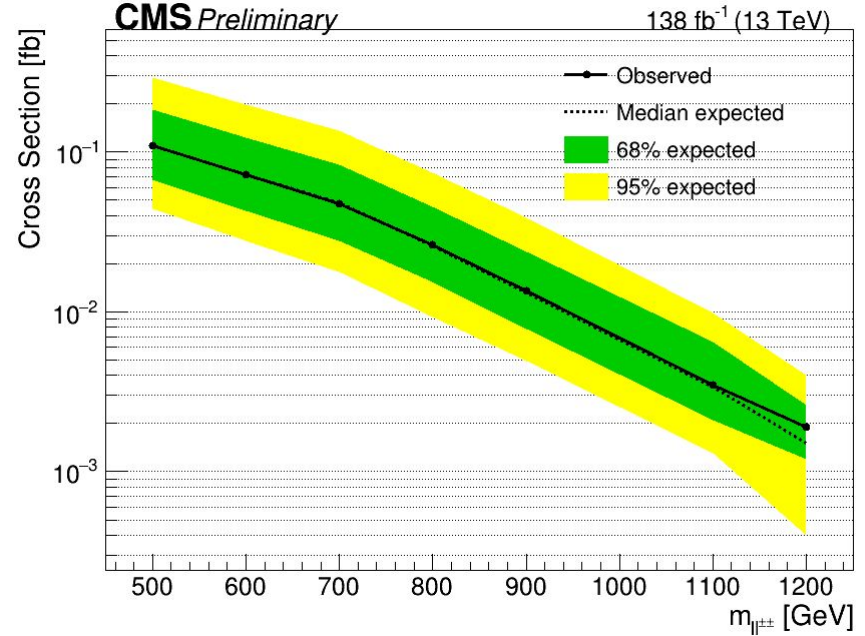
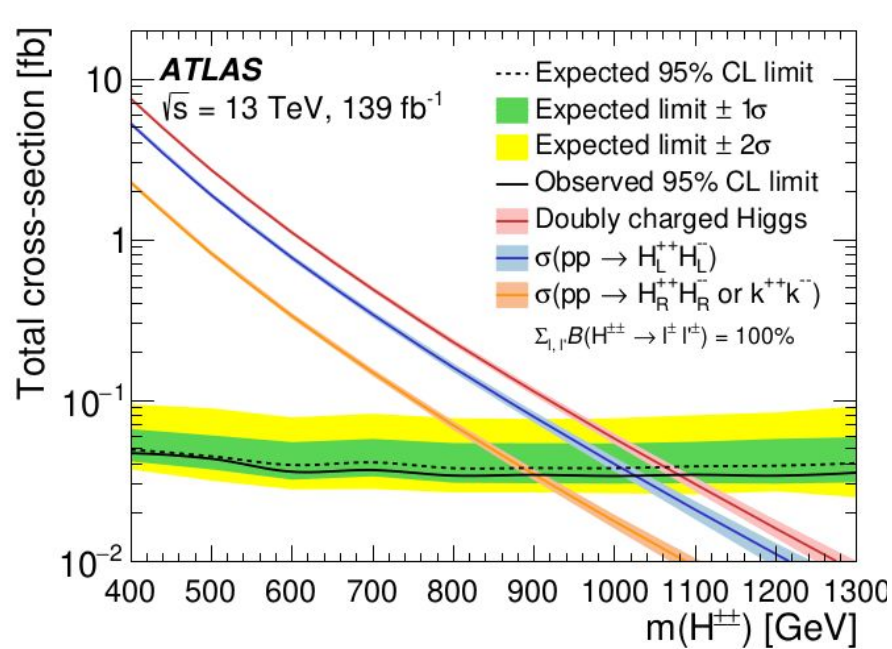


DCH preliminary limits

18 Sept 2026

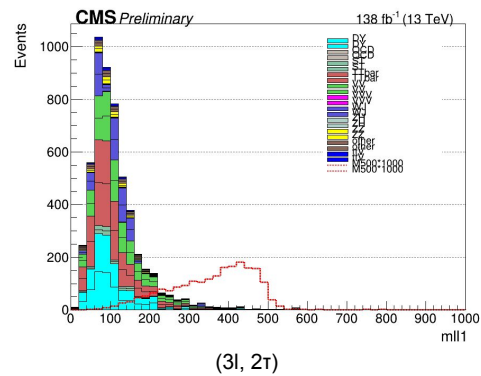
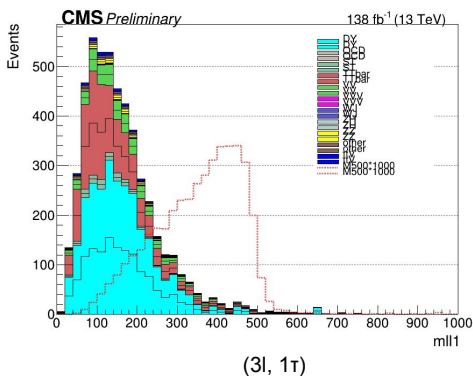
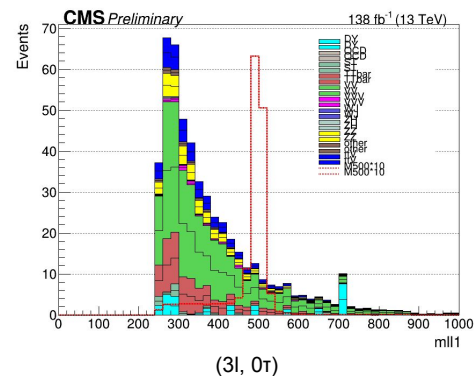
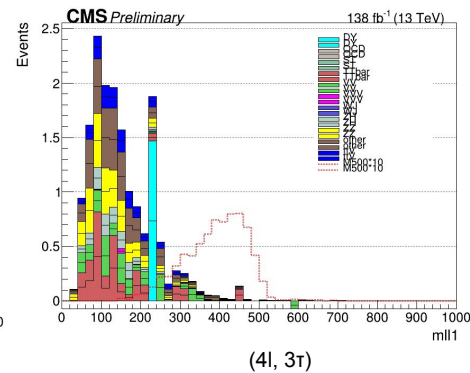
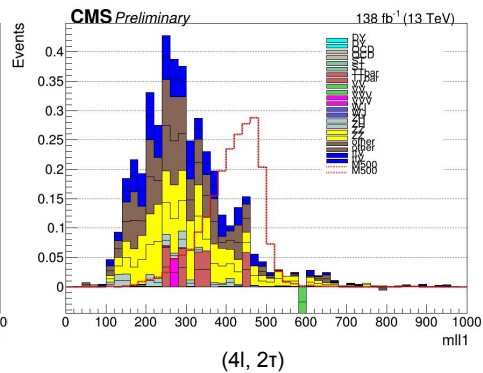
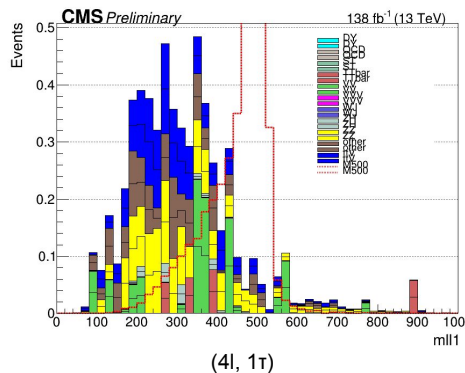
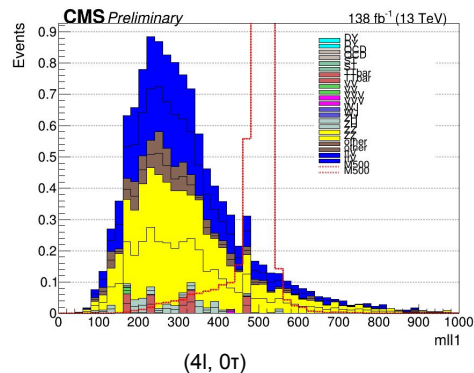
Limit comparison using ml1



We are assuming 1fb signal so the Y-axis is showing the crosssection limits in fb.

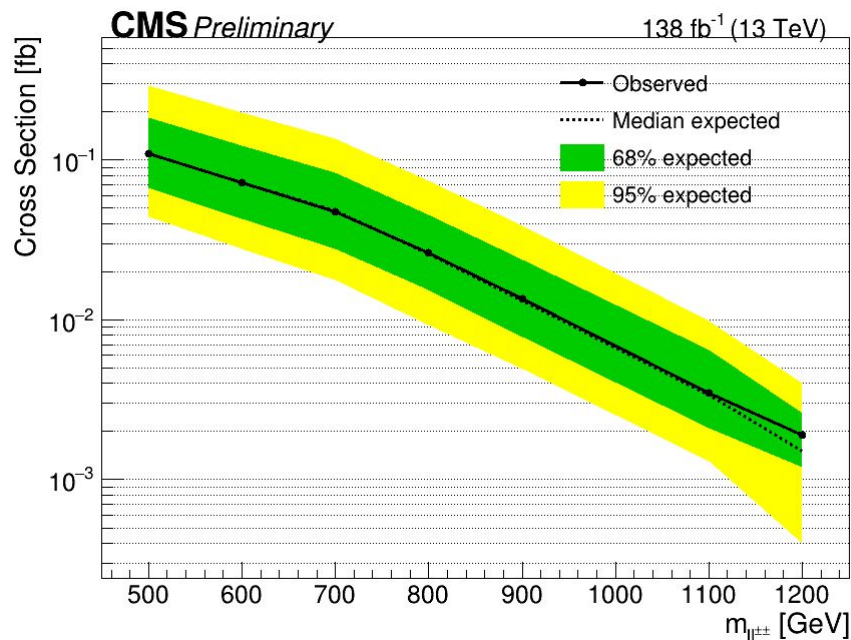
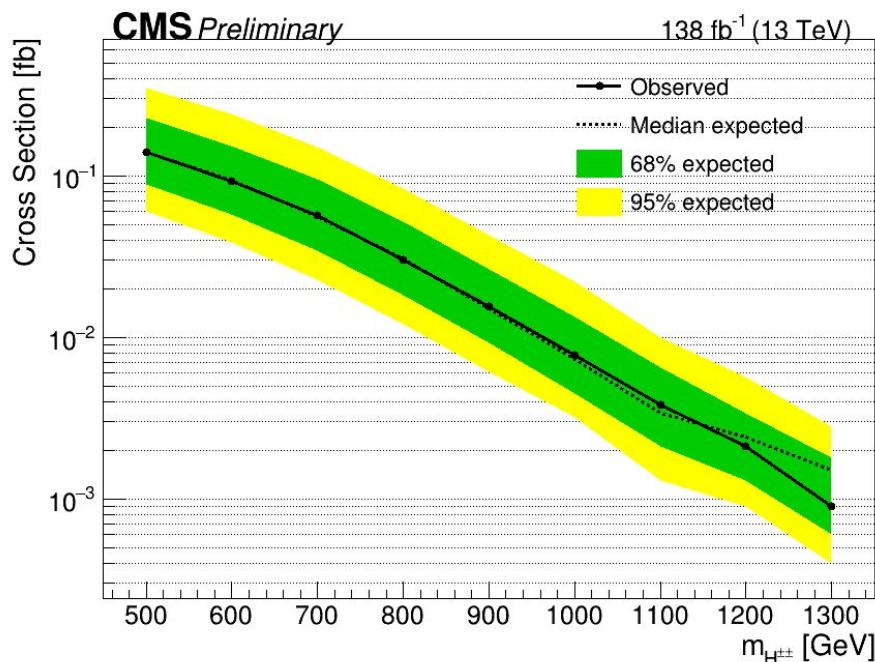
We get worse limits below $M_H < 700 \text{ GeV}$ but far better with higher masses. This is ml1 but is almost same with mDCH1 also!

ml1 templates for combine (sig = M500)



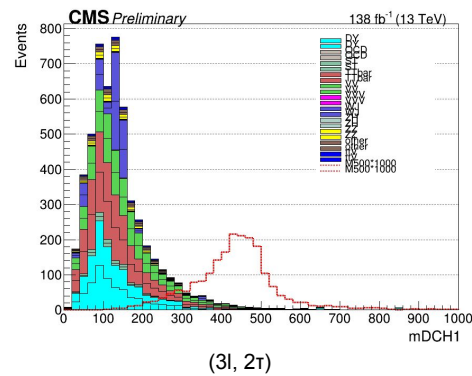
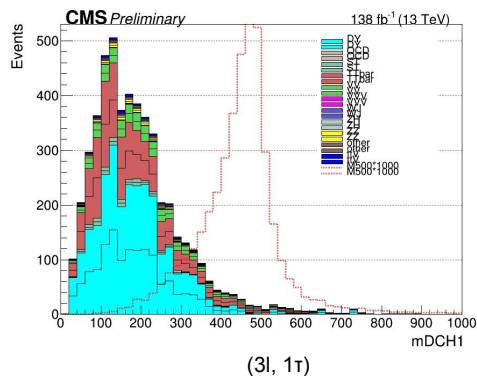
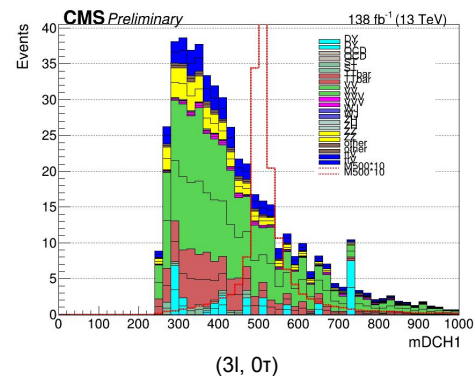
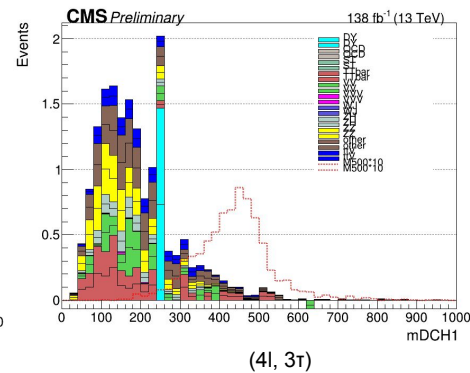
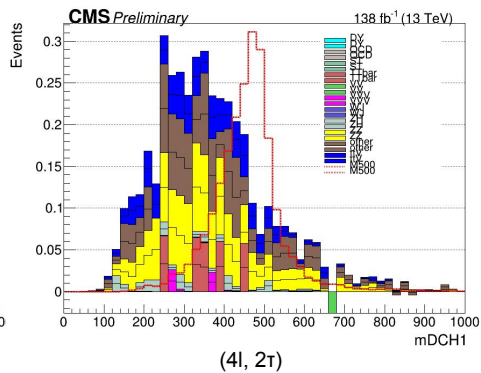
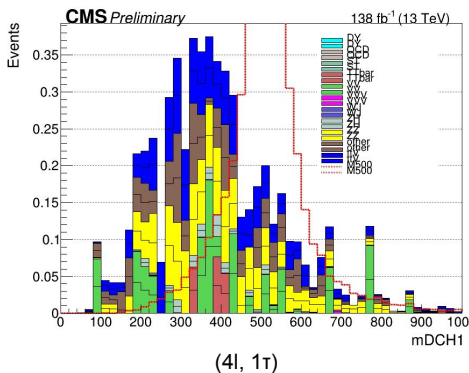
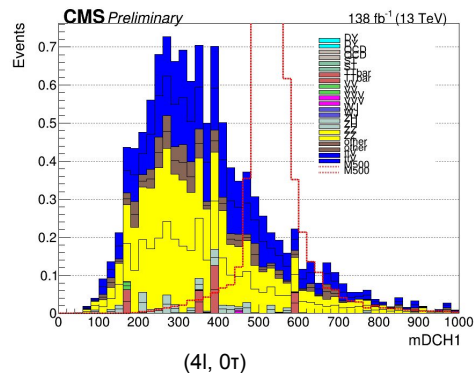
At lower masses, bkg pollution is significant for ml1.

Limit comparison using mll1 vs mDCH1



mDCH1 is giving almost same limits as mll1 and slightly worse at lower masses.

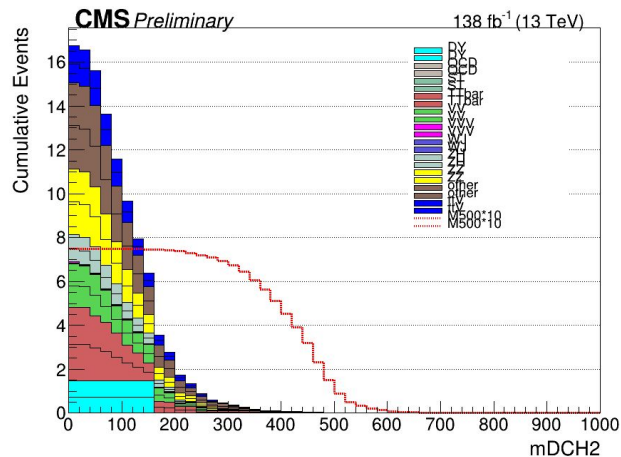
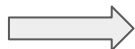
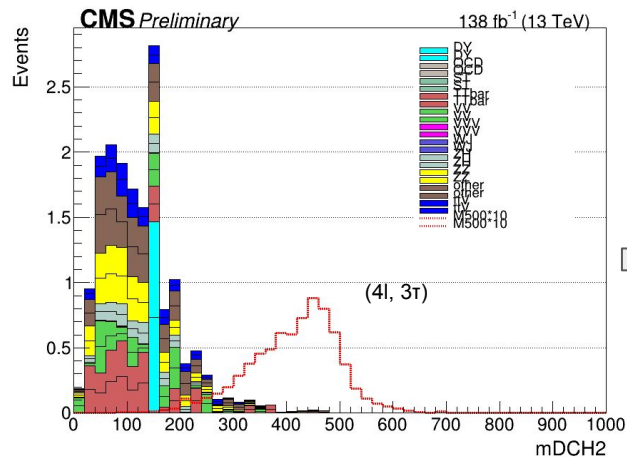
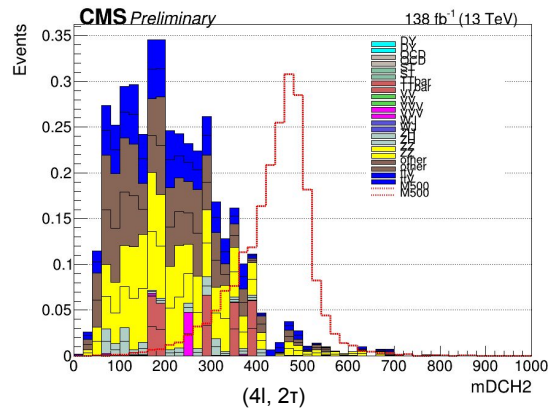
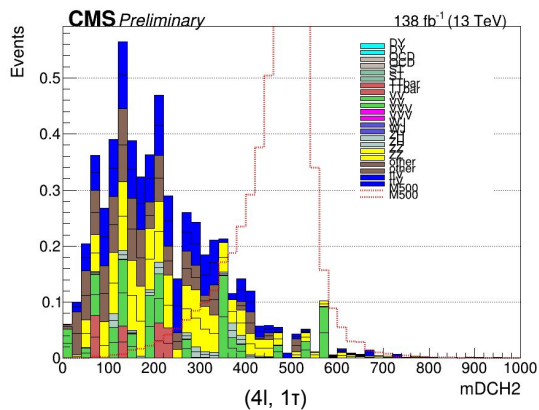
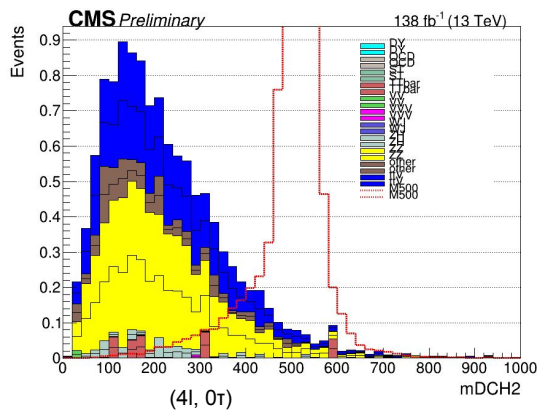
mDCH1 templates for combine (sig = M500)



Same for mDCH1.

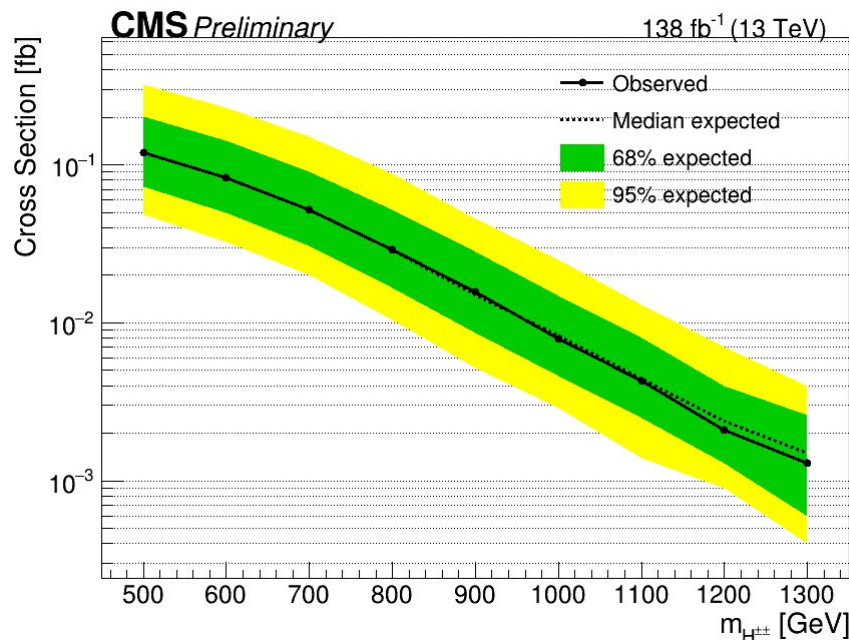
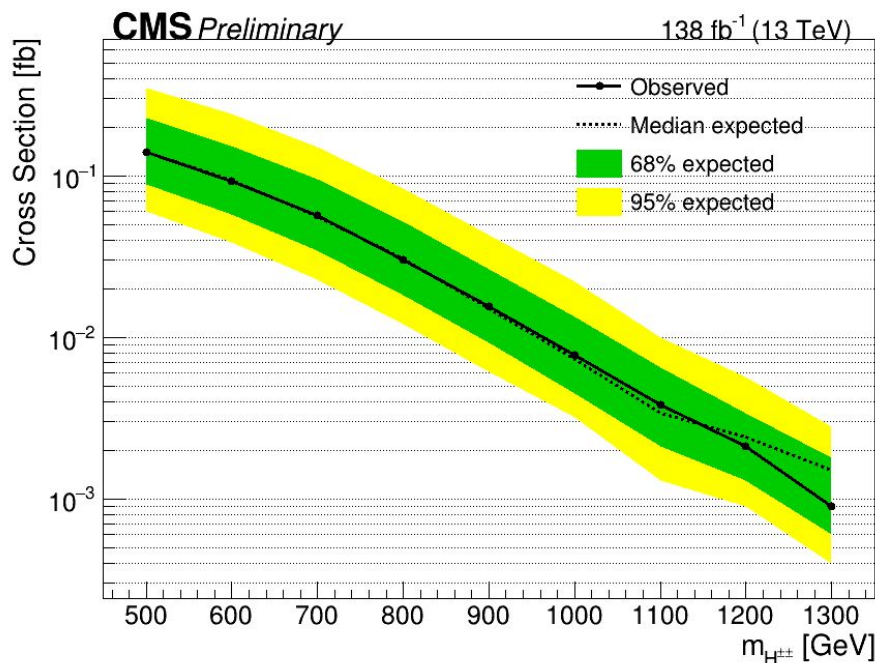
But our nu-algorithm
does a good job for
2nd pair. So we
should use it!

mDCH2 distribution



The signal falls off starting from 200 GeV. So we can cut at (mDCH2 > 200)

Limit comparison using mDCH1 before and after mDCH2 cut



mDCH1 is giving almost same limits as mll1 and slightly worse at lower masses.